

Part 3: Manual Gradient Descent

$$y = xm + b$$

$$\text{Given : } x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$m_0 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$b_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{Cost : } J = \frac{1}{2n} \|\hat{y} - y\|^2, n=2$$

$$\text{Gradient : } \nabla_m J = \frac{1}{n} X^T (\hat{y} - y)$$

$$\nabla_b J = \frac{1}{n} (\hat{y} - y)$$

$$\begin{aligned} \hat{y}_1 = X m_0 + b_0 &= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 \\ 16 \end{bmatrix} \end{aligned}$$

Gradient

$$e_1 = \hat{y}_1 - y = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\nabla_m J_1 = \frac{1}{2} X^T e_1 = \frac{1}{2} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 22.5 \\ 5 \end{bmatrix}$$

$$a = 0.01$$

$$m_1 = m_0 - a \nabla_m J_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 22.5 \\ 56.5 \end{bmatrix}$$

$$b_1 = b_0 - a \nabla_b J_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 0.5 \\ 5.5 \end{bmatrix} = \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix}$$

Iteration 2. By IGABE Plavide. Date / /

Prediction:

$$\hat{J}_2 = X m_1 + b_1 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.225 \\ 1.435 \end{bmatrix} + \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix}$$
$$= \begin{bmatrix} 3.08 \\ 10.35 \end{bmatrix} + \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix} = \begin{bmatrix} 4.075 \\ 11.295 \end{bmatrix}$$

$$\text{Error: } e = \begin{bmatrix} -0.925 \\ 5.295 \end{bmatrix}$$

$$\text{Gradients: } \nabla_m J_2 = \frac{1}{2} X^T e_2 = \frac{1}{2} \begin{bmatrix} -0.925 + 21.18 \\ -2.775 + 52.95 \end{bmatrix}$$
$$= \begin{bmatrix} 10.1275 \\ 25.0875 \end{bmatrix}$$

$$\nabla_b J_2 = \frac{1}{2} e_2 = \begin{bmatrix} -0.4625 \\ 2.6475 \end{bmatrix}$$

Updates:

$$m_2 = \begin{bmatrix} -1.225 \\ 1.435 \end{bmatrix} - 0.01 \begin{bmatrix} 10.1275 \\ 25.0875 \end{bmatrix} = \begin{bmatrix} -1.3263 \\ 1.1841 \end{bmatrix}$$

$$b_2 = \begin{bmatrix} 0.995 \\ 0.945 \end{bmatrix} - 0.01 \begin{bmatrix} -0.4625 \\ 2.6475 \end{bmatrix} = \begin{bmatrix} 0.9996 \\ 0.9185 \end{bmatrix}$$

Iteration 3. by Rugero Beulla Jeanne Françoise

$$\begin{aligned}\text{Prediction: } \hat{y} &= \begin{bmatrix} 1, 3 \\ 4, 10 \end{bmatrix} \times \begin{bmatrix} -1.3263, 1.1841 \end{bmatrix}^T + \begin{bmatrix} 0.9996, 0.9185 \end{bmatrix} \\ &= \begin{bmatrix} 2.226, 6.536 \end{bmatrix} + \begin{bmatrix} 0.9996, 0.9185 \end{bmatrix} \\ &= \begin{bmatrix} 3.2256, 7.4545 \end{bmatrix}\end{aligned}$$

$$\text{Error: } e = \begin{bmatrix} 3.2256, 7.4545 - 6 \end{bmatrix} = \begin{bmatrix} -1.7744, 1.4545 \end{bmatrix}$$

$$\begin{aligned}\text{Gradients: } \nabla_m J &= 1/2 \times X^T e \\ &= 1/2 \times \begin{bmatrix} (-1.7744 + 5.818), (-5.3238 + 14.545) \end{bmatrix} \\ &= \begin{bmatrix} 2.0218, 4.6109 \end{bmatrix}\end{aligned}$$

$$\nabla_b J = 1/2 \times \begin{bmatrix} -1.7744, 1.4545 \end{bmatrix} = \begin{bmatrix} -0.8872, 0.7273 \end{bmatrix}$$

$$\begin{aligned}\text{Update: } m_3 &= \begin{bmatrix} -1.3263, 1.1841 \end{bmatrix} - 0.01 \times \begin{bmatrix} 2.0218, 4.6109 \end{bmatrix} \\ &= \begin{bmatrix} -1.3465, 1.1380 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}b_3 &= \begin{bmatrix} 0.9996, 0.9185 \end{bmatrix} - 0.01 \times \begin{bmatrix} -0.8872, 0.7273 \end{bmatrix} \\ &= \begin{bmatrix} 1.0085, 0.9112 \end{bmatrix}\end{aligned}$$